Math 120 Spring 2020 Worksheet 11

1. Let $f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 1$.
   (a) Find all the critical points of $f(x, y)$.
   (b) Find all the local minima, local maxima, and saddle points of $f(x, y)$ if they exist.

2. Let $g(x, y) = x^{2012} + y^{2012}$. Find the critical points of $g(x, y)$ and classify them. (Hint: The second derivative test may fail at times. Notice that an even power of any number is non-negative, and thus $g(x, y) > 0$ except at the point $(0, 0)$.)

3. Find the dimensions of the largest rectangular box with volume 1000 cubic inches that has the smallest surface area. (Hint: Rectangular box has length, width, height. Find the equations of the volume and the surface area with respect to them.)